

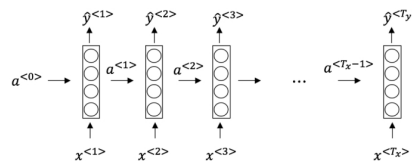
1. Suppose your training examples are sentences (sequences of words). Which of the following refers to the s^{th} word in the t^{th} training example?

1 point

- ☐ $x^{<t>}(s)$
- ☒ $x^{(s)}<s>$
- ☐ $x^{<s>}(t)$
- ☐ $x^{(s)}<t>$

2. Consider this RNN:

1 point

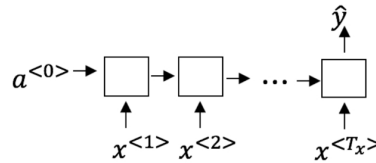


True/False: This specific type of architecture is appropriate when $T_x = T_y$

- ☒ False
- ☐ True

3. To which of these tasks would you apply a many-to-one RNN architecture? (Check all that apply).

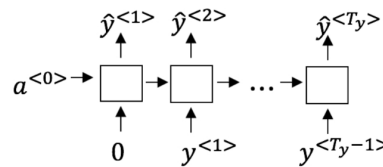
1 point



- ☐ Speech recognition (input an audio clip and output a transcript)
- ☒ Sentiment classification (input a piece of text and output a 0/1 to denote positive or negative sentiment)
- ☐ Image classification (input an image and output a label)
- ☒ Gender recognition from speech (input an audio clip and output a label indicating the speaker's gender)

4. You are training this RNN language model.

1 point

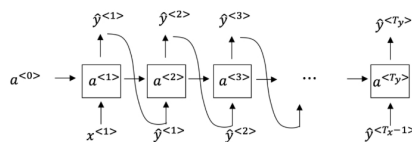


At the t^{th} time step, what is the RNN doing?

- ☐ Estimating $P(y^{<1>}, y^{<2>}, \dots, y^{<t-1>})$
- ☐ Estimating $P(y^{<t>})$
- ☒ Estimating $P(y^{<t>} | y^{<1>}, y^{<2>}, \dots, y^{<t-1>})$
- ☐ Estimating $P(y^{<t>} | y^{<1>}, y^{<2>}, \dots, y^{<t>})$

5. You have finished training a language model RNN and are using it to sample random sentences, as follows:

1 point



True/False: In this sample sentence, step 1 uses the probabilities output by the RNN to pick the highest probability word for that time-step. Then it passes the ground-truth word from the training set to the next time-step.

- ☒ True
- ☐ False

6. True/False: If you are training an RNN model, and find that your weights and activations are all taking on the value of NaN ("Not a Number") then you have an exploding gradient problem.

1 point

- ☐ False
- ☒ True

7. Suppose you are training an LSTM. You have an 80000 word vocabulary, and are using an LSTM with 800-dimensional activations $a^{<t>}$. What is the dimension of Γ_u at each time step?

1 point

- ☐ 8
- ☐ 80000
- ☒ 800
- ☐ 100

8. Sarah proposes to simplify the GRU by setting the update gate $\Gamma_u = 0$ (Always keep memory, never update).

1 point

Ashley proposes to simplify the GRU by setting the relevance gate $\Gamma_r = 1$ (Always consider previous memory relevant).

Which of these simplified models is more likely to work effectively without vanishing gradient problems when trained on very long input sequences?

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GRU (For Reference)

$$\tilde{c}^{(t)} = \tanh(W_c [\Gamma_r * c^{(t-1)}, x^{(t)}] + b_c)$$

$$\Gamma_u = \sigma(W_u [c^{(t-1)}, x^{(t)}] + b_u)$$

$$\Gamma_r = \sigma(W_r [c^{(t-1)}, x^{(t)}] + b_r)$$

$$c^{(t)} = \Gamma_u * \tilde{c}^{(t)} + (1 - \Gamma_u) * c^{(t-1)}$$

$$a^{(t)} = c^{(t)}$$

- ☐ Sarah's model (setting $\Gamma_u = 0$), because the cell state will never change, ensuring gradients are perfectly preserved.
- ☐ Sarah's model (setting $\Gamma_u = 0$), because removing the update gate simplifies the architecture, making it easier for the relevance gate Γ_r to learn long-term dependencies.
- ☐ Ashley's model (setting $\Gamma_r = 1$), because the relevance gate Γ_r is responsible for causing the vanishing gradient problem, so removing it solves the issue.
- ☒ Ashley's model (setting $\Gamma_r = 1$), because it retains the update gate Γ_u , which allows the network to control whether to keep or overwrite the memory cell.

9. Here are the equations for the GRU and the LSTM:

1 point

GRU	LSTM
$\tilde{c}^{<t>} = \tanh(W_c [\Gamma_r * c^{<t-1>}, x^{<t>}] + b_c)$	$\tilde{c}^{<t>} = \tanh(W_c [a^{<t-1>}, x^{<t>}] + b_c)$
$\Gamma_u = \sigma(W_u [c^{<t-1>}, x^{<t>}] + b_u)$	$\Gamma_u = \sigma(W_u [a^{<t-1>}, x^{<t>}] + b_u)$
$\Gamma_r = \sigma(W_r [c^{<t-1>}, x^{<t>}] + b_r)$	$\Gamma_f = \sigma(W_f [a^{<t-1>}, x^{<t>}] + b_f)$
$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + (1 - \Gamma_u) * c^{<t-1>}$	$\Gamma_o = \sigma(W_o [a^{<t-1>}, x^{<t>}] + b_o)$
$a^{<t>} = c^{<t>}$	$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + \Gamma_f * c^{<t-1>}$
	$a^{<t>} = \Gamma_o * \tanh c^{<t>}$

From these, we can see that the Update Gate and Forget Gate in the LSTM play a role similar to _____ and _____ in the GRU. What should go in the blanks?

- ☒ Γ_u and $1 - \Gamma_u$
- ☐ Γ_u and Γ_r
- ☐ $1 - \Gamma_u$ and Γ_u
- ☐ Γ_r and Γ_u

10. Your mood is heavily dependent on the current and past few days' weather. You've collected data for the past 365 days on the weather, which you represent as a sequence as $x^{<1>}, \dots, x^{<365>}$. You've also collected data on your mood, which you represent as $y^{<1>}, \dots, y^{<365>}$. You'd like to build a model to map from $x \rightarrow y$. Should you use a Unidirectional RNN or Bidirectional RNN for this problem?

1 point

- ☐ Unidirectional RNN, because the value of $y^{<t>}$ depends only on $x^{<t>}$, and not other days' weather.
- ☒ Unidirectional RNN, because the value of $y^{<t>}$ depends only on $x^{<1>}, \dots, x^{<t>}$, but not on $x^{<1>}, \dots, x^{<365>}$.
- ☐ Bidirectional RNN, because this allows backpropagation to compute more accurate gradients.
- ☐ Bidirectional RNN, because this allows the prediction of mood on day t to take into account more information.